Formlære: Mathematica

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The Axiomatic–Deductive Companion to Tractatus Formae

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Førsteutgåve
Sett og brote av forfattaren

Cross-references to the tractate are marked [TF …].

Every definition, postulate, and theorem listed in the Tractatus Formae appendix “Formell korrespondanse” finds its proof here.

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Preamble

This document provides the complete formal apparatus underlying *Tractatus Formae*. Where the tractate argues in natural language, this companion argues in set theory, topology, optimization theory, information theory, and dynamical systems. Every definition, postulate, and theorem referenced via [M: ...] in the tractate appears here with its full proof.

The reader is assumed to be familiar with real analysis, basic topology (compactness, continuity), and the calculus of variations.

Logical architecture. The theory rests on seven *postulates*, empirically falsifiable assumptions about the world, from which all theorems follow deductively. The postulates are:

1. **Proposition 1.4** (Explanation requirement)
2. **Postulate 2.3** (Multiple independent pressures)
3. **Postulate 2.6** (Conflicting gradients)
4. **Postulate 4.1** (Dynamic landscape)
5. **Postulate 5.1** (Geometric material signature)
6. **Postulate 7.1** (Substrate-independent navigation)
7. **Postulate 8.1** (Path-dependent fitness)

Numbering note. The *Mathematica* numbering (section.item) and the *Tractatus* numbering (decimal propositions) follow different architectures. The correspondence table in the appendix maps every formal element between the two systems.

Notation. We write M_C^{I} for the morphospace of class C (Def. 1.1), s_i for the i-th selection pressure (Def. 2.1), $\boldsymbol{s}$ for the selection-pressure vector (Def. 2.2), f for the fitness function (Def. 3.1), $\mathcal{L}$ for the fitness landscape (Def. 3.2), $\mathcal{N}$ for a navigator (Def. 5.1), and $\mathcal{P}$ for the Pareto front (Thm. 2.8). The gradient of a scalar field g is ∇g. Shannon entropy is H, mutual information is I.

1 Morphospace

Definition 1.1 (Morphospace)**.** Let C be a functional class of objects (e.g. chairs, knives, vessels) and let $x_1, x_2, \ldots, x_n$ be n measurable geometric properties that characterize objects in C (height, width, mass, curvature, symmetry, compactness, etc.).

For any object $a \in C$, its *attribute vector* is

$$\boldsymbol{x}(a) = \big(x_1(a),\ x_2(a),\ \ldots,\ x_n(a)\big) \in \mathbf{R}^n.$$

The *morphospace* of class C is the set of all attribute vectors satisfying the physical and functional constraints of C, equipped with the Euclidean metric $d(\boldsymbol{x}, \boldsymbol{y}) = \|\boldsymbol{x} - \boldsymbol{y}\|_2$.

Formally, let $\{g_j\}_{j=1}^m$ be a finite set of continuous constraint functions $g_j : \mathbf{R}^n \to \mathbf{R}$ encoding physical laws, material limits, and functional requirements. Then:

$$M_C^{\mathsf{I}} \;=\; \big\{\, \boldsymbol{x} \in \mathbf{R}^n \;\big|\; g_j(\boldsymbol{x}) \le 0 \;\; \forall\, j = 1, \ldots, m \,\big\}. \tag{1}$$

Since each g_j is continuous, M_C^{I} is a closed subset of $\mathbf{R}^n$. If additionally the property ranges are bounded (which holds for all physical objects), then M_C^{I} is *compact*. [TF 1.1, 1.11, 1.12]

Remark 1.2. Compactness of M_C^{I} is not a mathematical convenience but a physical fact: no chair has infinite height, negative mass, or unbounded curvature. Compactness guarantees that continuous functions on M_C^{I} attain their extrema (Weierstrass), which is essential for the existence of optima in the fitness landscape (§3).

Definition 1.3 (Region typology)**.** The full ambient space $\mathbf{R}^n$ partitions into three regions relative to the class C:

(i) **Busett (occupied) region** B: the support of the empirical measure μ on M_C^{I} induced by the population of realized objects. Formally, $B = \operatorname{supp}(\mu)$, which is closed in M_C^{I}.

(ii) **Open region** O:

$$O \;=\; M_C^{\mathsf{I}} \setminus B.$$

Positions that are physically feasible but historically unrealized.

(iii) **Forbidden region** F:

$$F \;=\; \mathbf{R}^n \setminus M_C^{\mathsf{I}}.$$

Positions that violate at least one constraint g_j.

Thus $M_C^{\mathsf{I}} = B \cup O$ with $B \cap O = \varnothing$, and $\mathbf{R}^n = M_C^{\mathsf{I}} \cup F = B \cup O \cup F$. [TF 1.13]

Proposition 1.4 (Explanation requirement)**.** If B contains more than one distinct position, then the fact that a particular object a occupies position $\boldsymbol{x}(a)$ and not some other feasible position $\boldsymbol{y} \in M_C^{\mathsf{I}}$ requires explanation.

Proof. Since M_C^{I} is a compact subset of $\mathbf{R}^n$ with $n \geq 1$ and B contains at least two distinct points, there exist at least two distinct feasible positions. By the principle of sufficient reason: if there were no cause favouring $\boldsymbol{x}(a)$ over $\boldsymbol{y}$, the occupation of $\boldsymbol{x}(a)$ would be arbitrary. But the empirical distribution over B is non-uniform (Postulate **??**), so the explanation must come from *selection pressures* that favour certain positions. □

[TF 1.14]

2 Selection Pressures

Definition 2.1 (Selection pressure)**.** A *selection pressure* on morphospace M_C^{I} is a continuous function

$$s_i : M_C^{\mathsf{I}} \longrightarrow \mathbf{R}$$

that assigns to each position $\boldsymbol{x}$ a real value $s_i(\boldsymbol{x})$ expressing how well $\boldsymbol{x}$ satisfies the i-th criterion. Higher values indicate better satisfaction. The gradient $\nabla\, s_i(\boldsymbol{x})$, where it exists, points in the direction of steepest improvement of the i-th criterion. [TF 2.11, 2.12]

Definition 2.2 (Selection-pressure vector)**.** Given k selection pressures $s_1, \ldots, s_k$ acting on M_C^{I}, the *selection-pressure vector* is the map

$$\boldsymbol{s} : M_C^{\mathsf{I}} \longrightarrow \mathbf{R}^k, \qquad \boldsymbol{s}(\boldsymbol{x}) = \big(s_1(\boldsymbol{x}),\ \ldots,\ s_k(\boldsymbol{x})\big).$$

Each component s_i represents an independent criterion. [TF 2.14]

Definition 2.3 (Functional delimitation)**.** Let $F_C : \mathbf{R}^n \to \{0, 1\}$ be the indicator function of the functional requirement:

$$F_C(\boldsymbol{x}) = \begin{cases} 1 & \text{if } \boldsymbol{x} \text{ satisfies the functional} \\ & \text{requirement of class } C, \\ 0 & \text{otherwise.} \end{cases}$$

Then $M_C^{\mathsf{I}} = \{\boldsymbol{x} \in \mathbf{R}^n : F_C(\boldsymbol{x}) = 1 \text{ and } g_j(\boldsymbol{x}) \leq 0\ \forall j\}$. The function defines the class and thereby the morphospace. Within M_C^{I}, $F_C \equiv 1$, so the function is *silent*: it delimits but does not differentiate. [TF 2.4]

Postulate 2.4 (Multiple independent pressures)**.** For every functional class C with morphospace M_C^{I}:

(a) $k \geq 2$: at least two selection pressures act on M_C^{I}.

(b) There exist at least two pressures s_i, s_j that are *statistically independent* with respect to the uniform measure on M_C^{I}: knowing $s_i(\boldsymbol{x})$

does not determine $s_j(\boldsymbol{x})$.

Falsification criterion. The postulate fails if there exists a class C in which a single selection pressure s_1 is sufficient to explain all observed variation in B, i.e., B is one-dimensional in the image of $\boldsymbol{s}$ and no second independent pressure can be identified. [TF 2.2]

Theorem 2.5 (Variation under constant function)**.** *Let C be a functional class with $k \geq 2$ selection pressures. Let s^f denote the functional pressure (how well the object serves its purpose). By Definition 2.3, all objects in C share the same function, so $s^f(\boldsymbol{x})$ is constant for all $\boldsymbol{x} \in M_C^{\mathsf{I}}$ that satisfy the functional requirement. Then:*

(i) s^f cannot explain any variation in form within C.

(ii) All observed form variation is attributable to the remaining pressures s^k ($k \neq f$).

Proof. (i) By Definition 2.3, every $\boldsymbol{x} \in B$ satisfies the functional criterion. Thus s^f is constant on B: $s^f(\boldsymbol{x}) = c$ for all $\boldsymbol{x} \in B$. A constant function has zero gradient: $\nabla s^f = \mathbf{0}$ everywhere on B. Therefore s^f induces no directional preference and cannot explain why any position is occupied rather than another.

(ii) By Postulate 2.4, $k \geq 2$ and at least one pressure s^j ($j \neq f$) is independent of s^f. Since s^f is constant, the observed variation in B (which is non-trivial since B contains more than one point) must be explained by the gradients of the remaining pressures s^j, $j \neq f$. □

[TF 2.21]

Postulate 2.6 (Conflicting gradients)**.** For at least one pair (s_i, s_j) of selection pressures, there exists a region $R \subseteq M_C^{\mathsf{I}}$ with $|R| > 0$ (positive measure) such that

$$\langle \nabla s_i(\boldsymbol{x}), \nabla s_j(\boldsymbol{x}) \rangle < 0 \qquad \forall \boldsymbol{x} \in R.$$

That is: improving one property worsens another.

Falsification criterion. The postulate fails if all gradient pairs are positively correlated everywhere in M_C^{I}, i.e., all pressures can be simultaneously improved at every point. [TF 2.3]

Theorem 2.7 (Compromise)**.** *Under Postulates 2.4 and 2.6, there exists no position $\boldsymbol{x}^* \in M_C^{\mathsf{I}}$ at which all selection pressures are simultaneously maximized. Every realized position $\boldsymbol{x} \in B$ is a* compromise.

Proof. Suppose for contradiction that $\boldsymbol{x}^*$ simultaneously maximizes all s_i. Then $\nabla s_i(\boldsymbol{x}^*) = \mathbf{0}$ for all $i = 1, \ldots, k$, so $\boldsymbol{x}^*$ is a critical point of each s_i individually. In particular, $\boldsymbol{x}^*$ is the global maximum of s_i on M_C^{I} (since we assumed it maximizes all pressures) and separately the global maximum of s_j.

Since M_C^{I} is compact and each s_i is continuous, global maxima exist (Weierstrass). Let $\boldsymbol{x}_i^* = \arg\max_{\boldsymbol{x} \in M_C^{\mathsf{I}}} s_i(\boldsymbol{x})$ and $\boldsymbol{x}_j^* = \arg\max_{\boldsymbol{x} \in M_C^{\mathsf{I}}} s_j(\boldsymbol{x})$. Our assumption requires $\boldsymbol{x}^* = \boldsymbol{x}_i^* = \boldsymbol{x}_j^*$.

We show these individual maxima must be distinct, contradicting the assumption. By Postulate 2.6, there exists a region $R \subseteq M_C^{\mathsf{I}}$ with positive measure where $\langle \nabla s_i, \nabla s_j \rangle < 0$.

If $\boldsymbol{x}_i^* = \boldsymbol{x}_j^*$, consider the behaviour of s_i and s_j near this common maximum. Near a non-degenerate maximum, the Hessian is negative definite, so the direction of steepest ascent to the maximum is unique from each nearby point. But in R, moving toward the maximum of s_i (in the direction ∇s_i) necessarily moves away from the maximum of s_j (since $\langle \nabla s_i, \nabla s_j \rangle < 0$). If R has positive measure, there is an open set of points from which s_i and s_j cannot be simultaneously increased. In particular, any path from R to the common point $\boldsymbol{x}^*$ cannot simultaneously ascend both s_i and s_j, which contradicts $\boldsymbol{x}^*$ being the maximum of both.

Therefore $\boldsymbol{x}_i^* \neq \boldsymbol{x}_j^*$, and no single point maximizes all pressures simultaneously. Every realized form balances competing demands: it is a compromise. □

[TF 2.31]

Theorem 2.8 (Multiple valid compromises)**.** *Under the conditions of Theorem 2.7, and given that B contains more than one point (Proposition 1.4), there exist multiple distinct compromises: different weightings of the pressures produce different optima.*

Proof. Let $\Phi_{\boldsymbol{w}} : \mathbf{R}^k \to \mathbf{R}$ be a monotonically increasing aggregation function parameterized by weight vector $\boldsymbol{w} = (w_1, \ldots, w_k) \in \Delta^{k-1}$ (the $(k-1)$-simplex). Define the $\boldsymbol{w}$-weighted fitness:

$$f_{\boldsymbol{w}}(\boldsymbol{x}) = \sum_{i=1}^{k} w_i \, s_i(\boldsymbol{x}).$$

Since M_C^{I} is compact and $f_{\boldsymbol{w}}$ is continuous, $f_{\boldsymbol{w}}$ attains its maximum at some $\boldsymbol{x}^*(\boldsymbol{w}) \in M_C^{\mathsf{I}}$.

For distinct weight vectors $\boldsymbol{w} \neq \boldsymbol{w}'$, the maxima generically differ: $\boldsymbol{x}^*(\boldsymbol{w}) \neq \boldsymbol{x}^*(\boldsymbol{w}')$. To see this, note that at $\boldsymbol{x}^*(\boldsymbol{w})$, the first-order condition requires

$$\sum_{i=1}^{k} w_i \, \nabla s_i(\boldsymbol{x}^*) = \mathbf{0}.$$

Under Postulate 2.6, the gradients ∇s_i are not positively proportional, so different $\boldsymbol{w}$ generically yield different critical points. Therefore the set of compromises $\{\boldsymbol{x}^*(\boldsymbol{w}) : \boldsymbol{w} \in \Delta^{k-1}\}$ contains more than one element, and this set traces out the *Pareto front* $\mathcal{P} \subseteq M_C^{\mathsf{I}}$.

The difference between two compromises under the same function is not a defect but reflects different balancings of the same forces. □

[TF 2.32]

3 Fitness Landscape

Definition 3.1 (Fitness function)**.** The *fitness function* (or *adaptation function*) is a continuous map

$$f : M_C^{\mathsf{I}} \longrightarrow \mathbf{R}, \quad f(\boldsymbol{x}) = \Phi\big(s_1(\boldsymbol{x}),\ \ldots,\ s_k(\boldsymbol{x})\big)$$

where $\Phi : \mathbf{R}^k \to \mathbf{R}$ is a smooth aggregation function, monotonically non-decreasing in each argument. The choice of Φ is not unique; different aggregations encode different relative weightings of the pressures. [TF 3.1]

Remark 3.2. In the simplest model Φ is linear: $f(\boldsymbol{x}) = \sum_i w_i\, s_i(\boldsymbol{x})$ with $w_i \geq 0, \sum w_i = 1$. But Φ may be nonlinear if pressures interact epistatically. [TF 3.11]

Definition 3.3 (Fitness landscape)**.** The *fitness landscape* is the graph of f over M_C^{I}:

$$\mathcal{L} \;=\; \big\{\, (\boldsymbol{x},\, f(\boldsymbol{x})) \;\big|\; \boldsymbol{x} \in M_C^{\mathsf{I}} \,\big\} \;\subseteq\; \mathbf{R}^{n+1}.$$

Local maxima of f are *peaks*. Local minima are *valleys*. Saddle points are *ridges* separating peaks. [TF 3.2]

Theorem 3.4 (Multiple peaks)**.** *Under Postulates 2.4 and 2.6, the fitness function f generically has at least two local maxima on M_C^{I}.*

Proof. Let s_i, s_j have conflicting gradients in a region R of positive measure (Postulate 2.6). Consider the one-parameter family of fitness functions

$$f_t(\boldsymbol{x}) = t\, s_i(\boldsymbol{x}) + (1-t)\, s_j(\boldsymbol{x}) \quad \text{for } t \in [0,1].$$

At $t = 1$, $f_1 = s_i$ attains its maximum at $\boldsymbol{x}_i^* = \arg\max s_i$. At $t = 0$, $f_0 = s_j$ attains its maximum at $\boldsymbol{x}_j^*$. By Theorem 2.7, $\boldsymbol{x}_i^* \neq \boldsymbol{x}_j^*$.

We verify the hypotheses of the mountain-pass lemma. The functional f_t is continuous on the compact set M_C^{I}, so it satisfies the Palais–Smale

condition (every sequence $\{x_k\}$ with $f_t(x_k)$ bounded and $\nabla f_t(x_k) \to 0$ has a convergent subsequence; this is automatic on a compact domain). For the mountain-pass geometry: let $c_0 = f_t(\boldsymbol{x}_i^*)$ and $c_1 = f_t(\boldsymbol{x}_j^*)$ be the values at the two endpoints. The gradient conflict in R implies that f_t is strictly below $\max(c_0, c_1)$ on a neighbourhood separating the two maxima, since moving from one peak toward the other requires simultaneously descending in at least one of s_i, s_j.

Define $\Gamma = \{\gamma \in C([0,1], M_C^{\mathsf{I}}) : \gamma(0) = \boldsymbol{x}_i^*,\ \gamma(1) = \boldsymbol{x}_j^*\}$. By the mountain-pass theorem, the minimax value

$$c = \inf_{\gamma \in \Gamma} \max_{s \in [0,1]} f_t(\gamma(s))$$

is a critical value of f_t, realized at a saddle point $\boldsymbol{x}_s$ with $f_t(\boldsymbol{x}_s) = c$. This saddle separates the basins of $\boldsymbol{x}_i^*$ and $\boldsymbol{x}_j^*$.

For the aggregated fitness $f = \Phi(\boldsymbol{s})$ with Φ monotone and smooth, the same structure persists: the basins of $\boldsymbol{x}_i^*$ and $\boldsymbol{x}_j^*$ in the individual pressures create distinct basins in f. Each basin of f is a compact connected subset of M_C^{I}; since f is continuous on each, it attains a maximum there (Weierstrass). Hence f has at least two local maxima on M_C^{I}. □

[TF 3.3]

Definition 3.5 (Style)**.** A *style* is a connected component of the occupied region B concentrated around a local maximum $\boldsymbol{x}^*$ of f:

$$S(\boldsymbol{x}^*, \varepsilon, \delta) \;=\; \left\{ \boldsymbol{x} \in B \;\middle|\; \|\boldsymbol{x} - \boldsymbol{x}^*\| < \varepsilon \text{ and } f(\boldsymbol{x}) > f(\boldsymbol{x}^*) - \delta \right\}$$

for suitable $\varepsilon > 0$, $\delta > 0$. A style is an attraction basin in the fitness landscape, restricted to realized objects. Styles are statistical phenomena, not categorical ones: boundaries are diffuse, and membership is graded. [TF 3.4, 1.2, 1.21]

Corollary 3.6 (Analyst-dependence of style decomposition)**.** *The decomposition of B into styles depends on the choice of aggregation function Φ and*

weights $\boldsymbol{w}$. Different analysts choosing different Φ may identify different numbers and boundaries of styles.

Proof. By Theorem 2.8, different weight vectors $\boldsymbol{w}$ produce different fitness functions $f_{\boldsymbol{w}}$ with different local maxima. Since styles are defined relative to local maxima of f (Definition 3.5), and f depends on Φ and $\boldsymbol{w}$, the style decomposition is a function of the analyst's choice. What is *not* analyst-dependent is the morphospace M_C^{I} itself, the selection pressures s_i, and the empirical distribution over B. The landscape is a construction; the pressures are real. □

[TF 3.41]

Definition 3.7 (Path (wandering))**.** A *formgiving history* (design trajectory) is a continuous path in morphospace:

$$\gamma : [0, T] \longrightarrow M_C^{\mathsf{I}}$$

with $\gamma(0) = \boldsymbol{x}^0$ (initial form) and $\gamma(T) = \boldsymbol{x}^T$ (final form). Continuity guarantees local increments: for every $\varepsilon > 0$ there exists $\delta > 0$ such that $|t' - t| < \delta$ implies $\|\gamma(t') - \gamma(t)\| < \varepsilon$. The path is *history-dependent*: the accessible region at time t depends on $\gamma\big([0, t]\big)$; this is formalized variationally in Postulate 8.1. [TF 3.6]

4 Dynamics

Postulate 4.1 (Dynamic landscape)**.** Selection pressures are time-dependent:

$$s_i(\boldsymbol{x},\, t) : M_C^{\mathsf{I}} \times \mathbf{R} \longrightarrow \mathbf{R}$$

and therefore the fitness landscape $\mathcal{L}(t)$ is a time-varying functional:

$$f(\boldsymbol{x},\, t) = \Phi\big(s_1(\boldsymbol{x}, t),\ \ldots,\ s_k(\boldsymbol{x}, t)\big).$$

Falsification criterion. The postulate fails if the fitness landscape for a class C can be shown to be topologically unchanged over a period in which the observed distribution of forms changes in a way not attributable to stochastic drift. [TF 4.1]

Theorem 4.2 (Punctuated equilibrium)**.** *Let $\boldsymbol{x}(t)$ be a form occupying a local maximum of $f(\cdot, t)$ at time t. Then:*

(i) ***Stability:*** *While $\partial f/\partial t$ is small relative to the curvature $\|\nabla^2 f\|$ at the peak, $\boldsymbol{x}(t)$ varies continuously and slowly (the style evolves gradually).*

(ii) ***Catastrophe:*** *When $\partial^2 f/\partial x^2$ changes sign at the peak (the maximum becomes a saddle point), $\boldsymbol{x}(t)$ undergoes a discontinuous jump to a new local maximum (style shift).*

Proof. (i) By the implicit function theorem, if $\boldsymbol{x}^*(t)$ is a non-degenerate local maximum of $f(\cdot, t)$, meaning the Hessian $H = \nabla^2 f(\boldsymbol{x}^*, t)$ is negative definite, then the map $t \mapsto \boldsymbol{x}^*(t)$ is smooth. The velocity of the peak is

$$\frac{d\boldsymbol{x}^*}{dt} = -H^{-1} \frac{\partial \nabla f}{\partial t}\bigg|_{\boldsymbol{x}^*}.$$

While $\partial f/\partial t$ is small and H remains negative definite (large curvature), $d\boldsymbol{x}^*/dt$ is small: the peak drifts slowly. Forms near this peak exhibit gradual stylistic evolution.

(ii) When a parameter t reaches a critical value t_c at which $\det(H) = 0$, the local maximum degenerates. In the generic case (fold catastrophe in the sense of Thom), two critical points, a maximum and a saddle, coalesce and annihilate. Beyond t_c, the original basin of attraction disappears, and trajectories that were stable near $\boldsymbol{x}^*(t)$ are redirected to a distant local maximum.

This is the mathematical structure underlying punctuated equilibrium: long periods of gradual drift punctuated by rapid transitions when the landscape topology changes. □

[TF 4.11, 4.12, 4.13]

Definition 4.3 (Landscape memory)**.** Every realized form $\boldsymbol{y} \in B$ modifies the future selection pressures via an *impact function* η_i:

$$s_i(\boldsymbol{x},\, t + dt) \;=\; s_i(\boldsymbol{x},\, t) + \eta_i(\boldsymbol{y},\, \boldsymbol{x})$$

where $\boldsymbol{y} = \gamma(t)$ is the currently realized form and $\eta_i : M_C^{\mathsf{I}} \times M_C^{\mathsf{I}} \to \mathbf{R}$ is continuous with $\eta_i(\boldsymbol{y}, \boldsymbol{y}) \geq 0$ (a realized form at least weakly reinforces its own position).

The landscape *remembers* past forms: current selection pressures are functions of the entire realized trajectory $\gamma([0, t])$, not just the current state. This creates co-evolutionary feedback between form and environment. [TF 4.2, 4.21, 4.22]

5 Navigator

Definition 5.1 (Navigator)**.** A *navigator* is a triple $\mathcal{N} = (G,\ \mu,\ \alpha)$ where:

(a) $G = G\big(\gamma([0,t])\big) \subseteq M_C^{\mathsf{I}}$ is a *goal set* (target region): a non-empty, possibly trajectory-dependent subset of morphospace that the navigator steers toward. The dependence on the trajectory history $\gamma([0,t])$ allows for goal revision, goal construction, and goal stacking across scales.

(b) $\mu : M_C^{\mathsf{I}} \to \mathbf{R}_{\geq 0}$ is a *distance function*: $\mu(\boldsymbol{x}) = d(\boldsymbol{x},\ G)$ for a metric d on M_C^{I}, with $\mu(\boldsymbol{x}) = 0 \iff \boldsymbol{x} \in G$.

(c) $\alpha : M_C^{\mathsf{I}} \to \mathbf{R}^n$ is a *steering field*: a vector field on M_C^{I} (in the ambient $\mathbf{R}^n$ sense) satisfying the *time-averaged descent condition*: for every t with $\mu(\gamma(t)) > 0$ and every characteristic timescale $\Delta > 0$,

$$\int_t^{t+\Delta} \langle \alpha(\gamma(\tau)), -\nabla\, \mu(\gamma(\tau)) \rangle\, d\tau \ > \ 0. \tag{2}$$

That is, α has a net component pointing toward decreasing distance to G, averaged over any interval, while permitting local exploratory deviations.

Three conditions are required: (a) the system has a goal state, (b) it can register distance from the goal, and (c) it can adjust toward the goal on average. No consciousness, intention, or nervous system is required. A thermostat, a gradient-descent algorithm, and a chemical reaction driven toward equilibrium are all navigators. [TF 5.1, 5.11]

Definition 5.2 (Cognitive light cone)**.** The *cognitive light cone* (boundary) of a navigator $\mathcal{N}$ at time t is the maximal region of morphospace–time in which $\mathcal{N}$ can represent states and steer:

$$\partial\mathcal{N} \ = \ \big\{\, (r,\ \tau) \in M_C^{\mathsf{I}} \times \mathbf{R} \ \big|\ \mathcal{N} \text{ can represent and steer toward states in } (r,\tau) \,\big\}. \tag{3}$$

The spatial extent determines what problems $\mathcal{N}$ can address; the temporal extent determines its planning horizon. [TF 5.2, 5.21]

Definition 5.3 (Intelligence)**.** The *intelligence* of a navigator $\mathcal{N}$ is its capacity to avoid local optima in the fitness landscape:

$$I(\mathcal{N}) \;\propto\; \max\{\, d\big(\gamma_{\mathcal{N}}(t),\; \boldsymbol{x}_{\text{local}}\big) \mid \gamma_{\mathcal{N}} \to \boldsymbol{x}_{\text{global}} \,\}$$

where $\boldsymbol{x}_{\text{local}}$ is a local optimum and $\boldsymbol{x}_{\text{global}}$ is a better alternative. Intelligence is the ability to follow detours: to accept temporarily lower fitness in order to reach a higher peak. It is scale-invariant: the definition applies equally to an enzyme avoiding a conformational trap and to a civilization investing in infrastructure. [TF 5.3, 5.31]

6 Empirical: Material Signatures

Postulate 6.1 (Geometric material signature)**.** For every material m used in functional class C, the conditional distribution of geometric properties given material is non-uniform:

$$P(\boldsymbol{x} \mid m) \neq \text{Uniform}(M_C^{\mathsf{I}}) \qquad \forall\, m.$$

That is, each material induces a characteristic geometric *profile*: a region of morphospace that it favours.

Falsification criterion. The postulate fails if the same geometric distribution is observed regardless of material, within the same functional class and the same set of other selection pressures. [TF 8.1]

Theorem 6.2 (Information theorem)**.** *Within a functional class C with at least two material categories, material carries strictly more mutual information with realized geometry than function does:*

$$\mathrm{I}(G;\, M) \;>\; \mathrm{I}(G;\, \mathrm{Func}) \;=\; 0.$$

Proof. Part 1: $\mathrm{I}(G;\, \mathrm{Func}) = 0$. By definition of a functional class, all objects in C share the same function. Therefore Func is a constant random variable within C: $\mathrm{H}(\mathrm{Func} \mid C) = 0$. By the data-processing inequality and the property that mutual information with a constant is zero:

$$\mathrm{I}(G;\, \mathrm{Func}) = \mathrm{H}(G) - \mathrm{H}(G \mid \mathrm{Func}) = \mathrm{H}(G) - \mathrm{H}(G) = 0.$$

Part 2: $\mathrm{I}(G;\, M) > 0$. By Postulate 6.1, $P(\boldsymbol{x} \mid m) \neq P(\boldsymbol{x})$ for at least one material m (the material-conditional distribution differs from the marginal). This is equivalent to G and M not being independent. By the fundamental property of mutual information:

$$\mathrm{I}(G;\, M) = 0 \iff G \perp\!\!\!\perp M.$$

Since $G \not\perp\!\!\!\perp M$ (by Postulate 6.1), we have $\mathrm{I}(G;\, M) > 0$.

Combining: $\mathrm{I}(G;\, M) > 0 = \mathrm{I}(G;\, \mathrm{Func})$. $\square$

[TF 8.11, 8.12, 8.13, 8.14]

Remark 6.3 (Empirical values from STOLAR dataset). In the dataset of $n = 93$ chairs from a Norwegian national museum (materials: mahogany, steel, laminate, beech; period 1280–2024):

$$\begin{aligned} \mathrm{I}(G;\, M) &= 0.382 \text{ bits}, \\ \mathrm{I}(G;\, \mathrm{Time}) &= 0.168 \text{ bits}, \\ \mathrm{I}(G;\, \mathrm{Geography}) &= 0.079 \text{ bits}, \\ \mathrm{I}(G;\, \mathrm{Func}) &= 0 \text{ bits (by definition)}. \end{aligned}$$

Material is the strongest predictor. This is the most concentrated empirical argument against the deterministic axiom "form follows function." [TF 2.61, 2.62, 8.11–8.18]

7 Substrate Independence and Emergence

Postulate 7.1 (Substrate-independent navigation)**.** The conditions (a), (b), (c) in Definition 5.1 can be satisfied by any physical system, biological, algorithmic, mechanical, or distributed, that:

(i) has states (positions in some configuration space),
(ii) can register distance from a target configuration, and
(iii) can adjust its state in response.

Navigation is a property of the system's *functional organization*, not of the material it is made from.

Falsification criterion. The postulate fails if it can be shown that at least one of conditions (a)–(c) necessarily requires a specific physical substrate (e.g., organic matter): that is, if there exist formgiving processes where a human navigator consistently produces forms that no algorithmic navigator can produce, *not* because of computational limits or sensory resolution, but because navigation itself requires organic substrate. [TF 5.12]

Theorem 7.2 (Position invariance of morphospace)**.** *The position $\boldsymbol{x} \in M_C^{\mathsf{I}}$ occupied by a realized object is determined by its measurable properties, not by the process that produced it. Two objects with identical attribute vectors occupy the same point in M_C^{I} regardless of whether they were made by hand, by machine, by algorithm, or by biological growth.*

Proof. By Definition 1.1, the attribute vector $\boldsymbol{x}(a)$ depends only on the measurable geometric properties $x_1(a), \ldots, x_n(a)$. These are intrinsic to the object a, not to its production history. If two objects a (handmade) and b (machine-made) have $x_i(a) = x_i(b)$ for all i, then $\boldsymbol{x}(a) = \boldsymbol{x}(b)$, hence they occupy the same point.

This does *not* mean the production process is irrelevant: it determines which region of M_C^{I} is accessible (the substrate constrains the trajectory, §8). But the position itself is substrate-invariant. □

[TF 1.4, 1.41, 5.12]

Remark 7.3. Theorem 7.2 establishes that the *output space* (morphospace) is substrate-invariant: position is determined by measurable properties, not production history. It does not establish that any specific navigational competency can be realized in multiple substrates. The latter is the content of Postulate 7.1, which remains an empirically motivated assumption. The framework provides substrate-neutral vocabulary; substrate independence of capacity is postulated, not proved.

Conjecture 7.4 (Emergent navigation)**.** *Let $\mathcal{N}^1, \mathcal{N}^2, \ldots, \mathcal{N}^S$ be a collection of sub-navigators, each with its own triple (G_j, μ_j, α_j). The collective system $\mathcal{N} = \{\mathcal{N}^1, \ldots, \mathcal{N}^S\}$ can realize a navigator triple (G, μ, α) where G, μ, and α are* not *functions of any single sub-navigator's triple, but emerge from the interaction dynamics.*

Formulation. Define the collective state as $\boldsymbol{X} = (\boldsymbol{x}^1, \ldots, \boldsymbol{x}^S) \in (M_C^{\mathsf{I}})^S$. Each sub-navigator $\mathcal{N}^j$ generates a flow $\dot{\boldsymbol{x}}^j = \alpha_j(\boldsymbol{x}^j)$ in its own morphospace. When sub-navigators interact through shared selection pressures, communication, or physical coupling, the collective dynamics become:

$$\dot{\boldsymbol{x}}^j = \alpha_j(\boldsymbol{x}^j) + \sum_{l \neq j} \phi_{jl}(\boldsymbol{x}^j, \boldsymbol{x}^l)$$

where ϕ_{jl} encodes the interaction between sub-navigators j and l.

The collective system exhibits a macroscopic goal G, distance μ, and steering α if there exists a coarse-graining map $\Pi : (M_C^{\mathsf{I}})^S \to M_C^{\mathsf{I}}$ such that the projected dynamics $\dot{\boldsymbol{y}} = \Pi_* \dot{\boldsymbol{X}}$ satisfy Definition 5.1 with (G, μ, α) that cannot be decomposed as functions of any single (G_j, μ_j, α_j).

The conditions on ϕ_{jl} under which such a map Π exists (e.g., sufficient coupling strength, symmetry, or mean-field limits) remain to be established.

Remark 7.5 (Empirical motivation). Synthetic organisms assembled from frog embryonic cells without evolutionary history navigate toward coherent morphological configurations. The navigation competency is not encoded in any single cell's genome but emerges from intercellular bioelectric signaling dynamics. This demonstrates that collective triples (G, μ, α)

arise as real emergent properties, not reducible to individual cell navigation. Similarly, the final form of a collaboratively designed object is not specified by any single participant but emerges from negotiation between them. These examples motivate the conjecture but do not constitute a mathematical proof of it. [TF 5.6, 5.61, 5.62]

Postulate 8.1 (Path-dependent fitness)**.** The fitness value of a realized form depends not only on its position $\boldsymbol{x} \in M_C^{\mathsf{I}}$ but on the entire path $\gamma : [0, T] \to M_C^{\mathsf{I}}$ taken to reach it. Define the *path fitness functional*:

$$S[\gamma] \;=\; \int_0^T g(\gamma(t),\; \dot{\gamma}(t),\; t)\, dt$$

where $g : M_C^{\mathsf{I}} \times \mathbf{R}^n \times \mathbf{R} \to \mathbf{R}$ is a Lagrangian that depends on the position $\gamma(t)$, velocity $\dot{\gamma}(t)$ (rate of change), and time t.

Interpretation. Two identical chairs occupying the same point in morphospace but with different production histories (e.g., original vs. copy) have different cultural value, different market position, and different potential to influence future designs. Position alone is insufficient; the path matters.

Falsification criterion. The postulate fails if identical forms with different production histories consistently have the same survival and reproduction rates in the design ecology. [TF 7.1]

Theorem 8.2 (Stationary path)**.** *If realized design trajectories are stationary points of $S[\gamma]$ (the conjecture $\delta S[\gamma] = 0$), then they satisfy the Euler–Lagrange equation:*

$$\frac{\partial g}{\partial \gamma} - \frac{d}{dt}\frac{\partial g}{\partial \dot{\gamma}} = 0.$$

Proof. This is a direct application of the calculus of variations. Let $\gamma_\varepsilon(t) = \gamma(t) + \varepsilon\, \eta(t)$ be a variation with $\eta(0) = \eta(T) = 0$. The stationarity condition is:

$$\left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0} S[\gamma_\varepsilon] = 0.$$

Computing:

$$\begin{aligned}\left.\frac{d}{d\varepsilon}\right|_0 S[\gamma_\varepsilon] &= \int_0^T \left(\frac{\partial g}{\partial \gamma}\cdot\eta + \frac{\partial g}{\partial \dot\gamma}\cdot\dot\eta\right) dt \\ &= \int_0^T \left(\frac{\partial g}{\partial \gamma} - \frac{d}{dt}\frac{\partial g}{\partial \dot\gamma}\right)\cdot\eta\, dt \\ &\quad + \left[\frac{\partial g}{\partial \dot\gamma}\cdot\eta\right]_0^T .\end{aligned}$$

The boundary term vanishes since $\eta(0) = \eta(T) = 0$. Since η is arbitrary, the fundamental lemma of the calculus of variations gives the Euler–Lagrange equation. □

[TF 7.11]

Theorem 8.3 (Path modifies landscape)**.** *The path-dependent fitness and landscape memory (Definition 4.3) imply that giving form is not finding the best position but finding the best* path*: the process modifies what outcomes are possible.*

Proof. By Definition 4.3, $s_i(\boldsymbol{x}, t+dt)$ depends on the realized form $\boldsymbol{y} = \gamma(t)$ via the impact function η_i. Since η_i is continuous on the compact set $M_C^{\mathsf{I}} \times M_C^{\mathsf{I}}$, it is integrable along any continuous path γ. The fitness landscape at time T is therefore a functional of the entire trajectory:

$$f(\boldsymbol{x}, T) = \Phi\left(s_1(\boldsymbol{x},\, 0) + \int_0^T \eta_1(\gamma(\tau), \boldsymbol{x})\, d\tau,\ \ldots\right).$$

The final landscape, and therefore the set of available optima, depends on the path taken. Two different trajectories γ and γ' arriving at the same final point $\boldsymbol{x}^T$ may face different future landscapes, different available optima, and different survival prospects. The process is not merely a means to the result; the process *modifies* what results are possible. □

[TF 7.12]

Theorem 8.4 (Spreading relation)**.** *Let $\sigma^2[\gamma]$ denote the variance of the form distribution around a given attractor in M_C^{I}, and let $\|\boldsymbol{s}\|$ denote the aggregate strength of selection pressures at that attractor. Then:*

$$\sigma^2 \;\propto\; \frac{1}{\|\boldsymbol{s}\|}.$$

Stronger selection produces narrower distributions (tighter styles); weaker selection produces broader distributions (more variation).

Proof. Model the stationary distribution of forms near a local maximum $\boldsymbol{x}^*$ as a Boltzmann-like distribution:

$$P(\boldsymbol{x}) \;\propto\; \exp\!\big(\beta\, f(\boldsymbol{x})\big)$$

where $\beta > 0$ controls the selection intensity (analogous to inverse temperature). This distributional form is an additional modeling assumption, not derived from the preceding postulates; it is motivated by the analogy between fitness-driven selection and statistical-mechanical equilibrium.

Near the maximum, expand f to second order:

$$f(\boldsymbol{x}) \approx f(\boldsymbol{x}^*) - \tfrac{1}{2}(\boldsymbol{x}-\boldsymbol{x}^*)^\top H(\boldsymbol{x}-\boldsymbol{x}^*)$$

where $H = -\nabla^2 f(\boldsymbol{x}^*)$ is positive definite (since $\boldsymbol{x}^*$ is a maximum). Then:

$$P(\boldsymbol{x}) \propto \exp\!\Big(-\tfrac{\beta}{2}(\boldsymbol{x}-\boldsymbol{x}^*)^\top H(\boldsymbol{x}-\boldsymbol{x}^*)\Big).$$

This is a multivariate Gaussian with covariance $\Sigma = (\beta H)^{-1}$. The variance along any axis is $\sigma_i^2 = (\beta\,\lambda_i)^{-1}$ where λ_i are the eigenvalues of H. Since H reflects the curvature of the aggregated selection pressures, $\lambda_i \propto \|\boldsymbol{s}\|$, giving $\sigma^2 \propto 1/\|\boldsymbol{s}\|$. □

[TF 7.2, 7.21]

9 Consequences

Theorem 9.1 (Provisional compromise)**.** *Every realized form is a* provisional *compromise: optimal for the selection pressures at the time of its creation, but generically suboptimal for any later configuration of pressures.*

Proof. By Theorem 2.7, every realized form $\boldsymbol{x} \in B$ is a compromise: it balances conflicting selection pressures but does not maximize all simultaneously. By Postulate 4.1, the pressures $s_i(\boldsymbol{x}, t)$ change over time.

Let $\boldsymbol{x}^*(\tau)$ be the local maximum of $f(\cdot, \tau)$ nearest to $\boldsymbol{x}$ at time τ of creation. At a later time $t > \tau$, the pressures have shifted: $\boldsymbol{s}(\cdot, t) \neq \boldsymbol{s}(\cdot, \tau)$. The point $\boldsymbol{x}^*(\tau)$ is generically *not* a critical point of $f(\cdot, t)$, since

$$\nabla f(\boldsymbol{x}^*(\tau),\, t) = \int_\tau^t \frac{\partial}{\partial t'} \nabla f(\boldsymbol{x}^*(\tau),\, t')\, dt' \neq \mathbf{0}$$

unless the integral vanishes identically (a non-generic condition). Therefore the form that was optimal at τ is no longer optimal at t. No form is final; every object is a snapshot of a dynamic system. □

[TF 7.4]

Theorem 9.2 (Durable traditions)**.** *Traditions with more independent adaptive mechanisms (more selection pressures addressed by competent sub-navigators) are more robust to landscape changes.*

Proof. Let a tradition $\mathcal{T}$ be a population of navigators addressing $k_\mathcal{T}$ distinct selection pressures through independent adaptive mechanisms. When the landscape shifts (Postulate 4.1), typically only a subset of pressures change simultaneously.

A tradition with $k_\mathcal{T}$ independent mechanisms can respond to a shift in pressure s_i without disrupting its response to pressures s_j ($j \neq i$), because the mechanisms are independent. Its adaptation requires adjusting only $\Delta k \leq k_\mathcal{T}$ mechanisms.

Compare traditions $\mathcal{T}_1$ with k_1 mechanisms and $\mathcal{T}_2$ with $k_2 > k_1$ mechanisms. Under a landscape shift affecting one pressure, $\mathcal{T}_2$ adjusts only $\Delta k/k_2 < \Delta k/k_1$ of its total adaptive capacity. The larger repertoire provides redundancy: a craft tradition that combines material understanding, cultural sensitivity, and individual innovation adapts better than one relying on a single mechanism. □

[TF 7.5]

Theorem 9.3 (Falsifiability of the framework)**.** *The framework of Tractatus Formae is falsifiable: if a single selection pressure s_1 is sufficient to explain all observed form variation within a functional class C, then Postulate 2.4 fails, and with it Theorems 2.5, 2.7, 2.8, 3.4, Corollary 3.6, and Theorem 9.1.*

Proof. The deductive chain is:

1. Postulate 2.4 (multiple independent pressures) $\implies$ Theorem 2.5 (variation not explained by function).
2. Postulates 2.4 + 2.6 (conflicting gradients) $\implies$ Theorem 2.7 (compromise) $\implies$ Theorem 2.8 (multiple compromises).
3. Theorems 2.7 + 2.8 $\implies$ Theorem 3.4 (multiple peaks) $\implies$ Definition 3.5 (styles exist).
4. Theorem 3.4 + Definition 3.5 $\implies$ Corollary 3.6 (analyst dependence).
5. Postulate 4.1 (dynamic landscape) + Theorem 2.7 $\implies$ Theorem 9.1 (provisional compromise).

If any of the seven postulates can be empirically disproven, the corresponding branch of the deductive tree collapses. The framework is valid precisely because it can fail. The deterministic axiom "form follows function" was unfalsifiable and therefore not a scientific claim. This framework makes quantitative predictions (Theorem 8.4, Postulate 6.1) that can be tested against data.

That the text *can* be felled is the reason it is valid. □

[TF 7.7]

Appendix: Correspondence Table

The following table maps every formal element in this document to its counterpart in *Tractatus Formae*.

Mathematica	Type	TF
Def. 1.1 (Morphospace)	Def.	1.1
Def. 1.2 (Region typology)	Def.	1.13
Prop. 1.4 (Explanation)	Prop.	1.14
Def. 2.1 (Selection pressure)	Def.	2.1
Def. 2.2 (Pressure vector)	Def.	2.14
Def. 2.3 (Functional delim.)	Def.	2.4
Post. 2.4 (Multiple pressures)	Post.	2.2
Thm. 2.5 (Variation under fn.)	Thm.	2.21
Post. 2.6 (Conflicting grad.)	Post.	2.3
Thm. 2.7 (Compromise)	Thm.	2.31
Thm. 2.8 (Multiple comp.)	Thm.	2.32
Def. 3.1 (Fitness function)	Def.	3.1
Def. 3.2 (Fitness landscape)	Def.	3.2
Thm. 3.3 (Multiple peaks)	Thm.	3.3
Def. 3.4 (Style)	Def.	3.4
Cor. 3.5 (Analyst dependence)	Cor.	3.41
Def. 3.6 (Path / wandering)	Def.	3.6
Post. 4.1 (Dynamic landscape)	Post.	4.1
Thm. 4.2 (Punctuated equil.)	Thm.	4.11
Def. 4.3 (Landscape memory)	Def.	4.2
Def. 5.1 (Navigator)	Def.	5.1
Def. 5.2 (Cognitive light cone)	Def.	5.2
Def. 5.3 (Intelligence)	Def.	5.3
Post. 6.1 (Geometric signature)	Post.	8.1
Thm. 6.2 (Information theorem)	Thm.	8.11
Post. 7.1 (Substrate indep.)	Post.	5.12
Thm. 7.2 (Position invariance)	Thm.	1.4

Mathematica	Type	TF
Conj. 7.3 (Emergent navig.)	Conj.	5.62
Post. 8.1 (Path-dep. fitness)	Post.	7.1
Thm. 8.2 (Stationary path)	Thm.	7.11
Thm. 8.3 (Path modifies land.)	Thm.	7.12
Thm. 8.4 (Spreading relation)	Thm.	7.2
Thm. 9.1 (Provisional comp.)	Thm.	7.4
Thm. 9.2 (Durable traditions)	Thm.	7.5
Thm. 9.3 (Falsifiability)	Thm.	7.7

Appendix: Logical Dependency Graph

The deductive structure of the theory, showing which results depend on which postulates and earlier theorems:

```
              +-- Def 1.1 (Morphospace)
              |    +-- Def 1.2 (Regions)
              |        +-- Prop 1.4 (Explanation)
              |
POST 2.4 ---+-- Def 2.3 (Functional delim.)
(k >= 2)    |    +-- Thm 2.5 (Var /= function)
              |
              |         +-- Thm 2.7 (Compromise)
POST 2.6 ---+---------+
(conflict)  |         +-- Thm 2.8 (Mult. comp.)
              |              |
              |        Thm 3.3 (Multiple peaks)
              |         +-- Def 3.4 (Style)
              |         +-- Cor 3.5 (Analyst dep.)
              |
POST 4.1 ---+-- Thm 4.2 (Punct. equil.)
(dynamic)   |   +-- Def 4.3 (Memory)
              |
              +-- Thm 9.1 (Provis. comp.)
              |       [from 2.7 + 4.1]
              +-- Thm 9.2 (Durable trad.)
                      [from 9.1 + 5.1]

POST 6.1 ---- Thm 6.2 (Information thm)
(material)

POST 7.1 ---- Thm 7.2 (Pos. invariance)
(substr.)      +-- Conj 7.3 (Emergence)

POST 8.1 ---+-- Thm 8.2 (Stat. path / E-L)
(path-dep.)  +-- Thm 8.3 (Path mod. land.)
             +-- Thm 8.4 (Spreading rel.)

Thm 9.3 (Falsifiability)
  [meta: Post 2.4 fails ->
    2.5, 2.7, 2.8, 3.3, 9.1 fall]
```

Summary of formal inventory:

- 12 definitions (Def. 1.1–5.3)
- 7 postulates (Post. 2.4, 2.6, 4.1, 6.1, 7.1, 8.1 + the explanation principle Prop. 1.4)

- 11 theorems (Thm. 2.5–9.3)
- 1 corollary (Cor. 3.5)
- 1 conjecture (Conj. 7.3)

This document is the formal companion to

Tractatus Formae

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